LECTURE 2.4 · SHIP AND OCEAN SYSTEMS DESIGN

Preliminary Power Estimate

Papanikolaou (2026) §2.14, §5.1 · Session W2.4 · Lecture 35 min + Exercise 2.4 45 min

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From Resistance to Installed Power

The first power sizes the machinery weight Wm of the design equation, the engine room, and the fuel or battery. A preliminary estimate is enough now (§2.14).

RT
total resistance [kN]

→

PE = RT·V
effective power

→

PD = PE/ηD
delivered, ηD = η0·ηH·ηR

→

PB = PD/ηS
brake power, ηS ≈ 0.99

→

SMCR = PB·1.15/0.90
15% sea margin, engine at 90%

ηH = (1 − t)/(1 − w): hull efficiency from wake w and thrust deduction t; η0 open-water propeller efficiency; ηR relative rotative efficiency. **In the baseline design η0 is a fixed assumption, 0.60;** the optimum propeller is selected in SP02.

> **Three routes in §2.14:** (a) the Admiralty formula and its variants · (b) diagrams, MAN B&W SMCR = f(TEU, V) and Harvald · (c) resistance methods, Holtrop–Mennen and Hollenbach (footnote 34).

The Admiralty Formula: One Constant from a Parent

$$C = \Delta^{2/3} \cdot V^3 / P \quad (2.104)$$

Δ in t, V in kn, P in kW. Typical C 400–600; higher means a more efficient ship.

Valid only for a parent of similar form, length and speed: the same Reynolds and Froude numbers. Variants: Völker (2.105), Heickel (2.106).

Reference ship, 14 kn	P [kW]
C = 400 (Exercise 1.5)	3,832
C = 500	3,065
C = 600	2,554

$\Delta^{2/3} = 558.6$, $V^3 = 2{,}744$

C is the whole answer. Power is inversely proportional to C: a C 20% lower means 25% more power. In your project, take it from a parent ship in your D2 database, not from the range.

Holtrop–Mennen: Resistance in Components

$$RT = RF(1 + k1) + RAPP + RW + RB + RTR + RA$$

Inputs: LWL, B, T, ∇, lcb, CM, CWP, bulb, transom, appendages, stern shape, propeller diameter

Outputs: each component, PE, and the propulsion factors w, t, ηR

A regression on model tests: use it inside its data range only.

Reference ship, 14 kn	kN	Share
Friction with form, RF(1 + k1)	174.4	65%
Wave-making, RW	49.7	19%
Model–ship correlation, RA	42.3	16%
Appendages, RAPP	1.9	1%
Total, RT	**268.3**	

PE 1,932 kW · w 0.305, t 0.186, ηR 1.013, η0 0.60 assumed → ηD 0.712 · PB 2,739 kW · **SMCR 3,500 kW**

Two Methods, One Ship: Power Against Speed

V [kn]	Fn (LWL)	RW share	SMCR, Holtrop–Mennen [kW]	Admiralty, C 400 [kW]	Implied C
12	0.175	8%	1,972	2,413	489
13	0.189	11%	2,596	3,068	473
14	0.204	19%	3,500	3,832	438
15	0.218	22%	4,454	4,713	423
16	0.233	33%	6,248	5,720	366

The speed exponent

P ∝ V^n: n is 3.4 at 12–13 kn and 5.2 at 15–16 kn as the waves grow. The Admiralty formula keeps n = 3.

The level

With C = 400 Lecture 1.5 was 9% high at 14 kn. A parent ship fixes the level; Holtrop–Mennen gives the shape.

Choosing the Machinery: §5.1

Twelve selection criteria

Acquisition cost	Safety in operation
Weight	Space
Specific fuel consumption	Fuel type and cost
Emissions (SOx, NOx, CO_2)	Repair cost
Maintenance cost	Maneuverability
Vibrations	Automation

EEDI ≈ Σ P·CF·SFC / (Capacity · Vref), g CO_2 per t·nm (5.1)
Container ships: Capacity = 70% DWT

Required = (1 − x)·174.22·DWT^−0.201 (5.2–5.3)

CF: HFO 3.114, LNG 2.750 t CO_2 per t fuel (Table 5.1)

Installed power sits in the numerator: for the Express, minimum power is also the EEDI strategy. The Feeder’s electric propulsion is non-conventional, so no EEDI is required; in normal service it runs on batteries (generators for the mainland transit and emergencies).

The Story in Four Lines

1 · Why. Power sizes machinery weight, engine room and fuel or battery: a preliminary estimate is enough now.

2 · Admiralty. One constant from a parent of the same size and speed; no parent, no answer.

3 · Holtrop–Mennen. Resistance in components plus w, t, ηR; check its data range.

4 · Machinery. Twelve criteria and the EEDI: installed power drives both.

Next: Exercise 2.4 on the reference ship with Tool 2.4 (slide 8). Week 3 (W3.1): weights, centroids and the displacement check, §2.15–2.16.

EXERCISE 2.4 · INDIVIDUAL · 45 MIN · PRACTICE FOR D3

Power for the Reference Ship, Two Ways

The reference ship, the 10,000 t general cargo ship: Δ 13,200.7 t, 124.8 × 20.0 × 6.89 m, CB 0.749, 14 kn. Tool 2.4 opens on it; keep slides 3–5 closed until step 4.

1 Admiralty · 8 min

C 400 (Lecture 1.5): power at 12, 14 and 16 kn. Then the C implied by an SMCR of 3,500 kW at 14 kn.

2 Holtrop–Mennen · 10 min

Tool 2.4 at 14 kn: RT and its parts, PE, w, t, ηR, ηD, PB and SMCR (η0 0.60, 15% sea margin, 90% MCR).

3 New input: B 21 m · 10 min

Same L, CB and Δ, so T = 6.56 m and the propeller 0.70 T = 4.59 m. New RT, w, t, ηD and SMCR at 14 kn.

4 Compare · 7 min

Admiralty against Holtrop–Mennen and the implied C, both hulls. What did 1 m of beam cost in power, and what did it buy (Lecture 2.3a, slide 5)?

Check: the worked answer is shown at the end (Exercise Answers – Week 2). Not graded.
Your project: your group applies this to its own ship outside class and completes D3 (see your Project Scope Sheet).